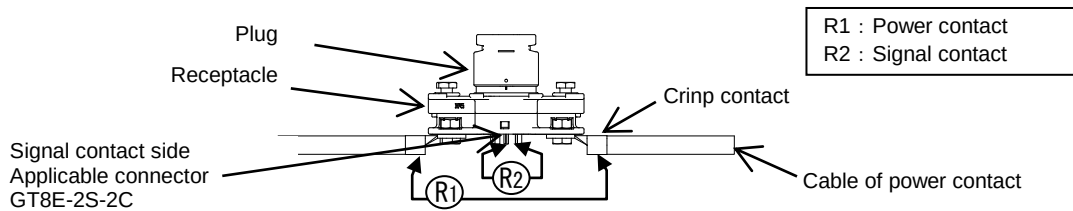
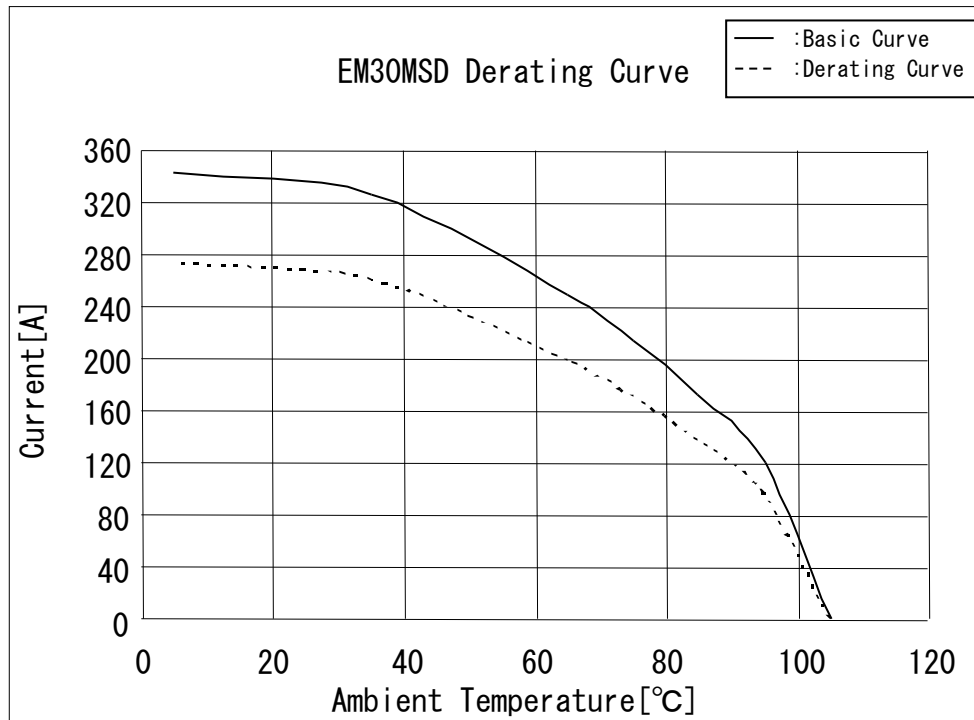


APPLICABLE STANDARD		TÜV approved (J50385364). UL approved (E474564).															
Rating	Operating Temperature Range	-40 °C to +105 °C ⁽⁵⁾	Storage Temperature Range	-10°C to +60°C													
	Voltage	Power Contact : AC/DC 1500 V Signal Contact : AC/DC 250 V															
	Current	Power Contact : 200 A ⁽³⁾ Signal Contact : 1 A	Applicable Cable	100 mm ² min (AWG#4/0 min)													
SPECIFICATIONS																	
ITEM		TEST METHOD		REQUIREMENTS	QT AT												
CONSTRUCTION																	
General Examination		Examined visually and with a measuring instrument.		According to the drawing.	X X												
Marking		Confirmed visually.			X X												
ELECTRICAL CHARACTERISTICS																	
Contact Resistance ⁽²⁾	Measured at DC 1A.(Power contact)		0.5 mΩ MAX.	X	X												
	Measured at DC 1A.(Signal contact include GT8E-2S-2C)		90 mΩ MAX.	X	X												
Insulation Resistance	Measured at 500 V DC.		5000 MΩ MIN.	X	X												
Voltage Proof	4500 V AC applied for 1 min. (Power contact)		No breakdown.	X	X												
	750 V AC applied for 1 min. (Signal contact)			X	X												
MECHANICAL CHARACTERISTICS																	
Mating and Unmating Forces	Measured with an applicable connector. Without locking device.		Mating and unmating forces: 100 N MAX. (Between EM30MSD Plug and Receptacle)	X	—												
Mechanical Operation	Mated and unmated 50 times. (Between EM30MSD Plug and Receptacle)		Contact resistance: 0.75 mΩ MAX. (Power contact) Contact resistance: 150 mΩ MAX. (Signal contact include GT8E-2S-2C)	X	—												
	Mated and unmated 30 times. (Between EM30MSD Receptacle and GT8E-2S-2C)		Contact resistance : 150 mΩ MAX (Signal contact include GT8E-2S-2C)	X	—												
Vibration 1	Frequency: 10 Hz to 55 to 10 Hz every cycle (5 min per cycle) Single amplitude: 0.75 mm Performed over 10 cycles in each of three mutually perpendicular directions.		1) No electrical discontinuity of more than 10 μs. 2) No damage, cracks or looseness of parts.	X	—												
Vibration 2 (Random) (ISO16750-3 / JASO D 014-3)	Frequency : 10 to 2000 (Hz), Acceleration spectrum density : 57.9 m/s ² , At 8 h, for 3 directions.		1) No electrical discontinuity of more than 10 μs. 2) No damage, cracks or looseness of parts.	X	—												
Shock	Acceleration: 490 m/s ² , Half sine wave pulses of 11 ms. Performed 3 times in each of three mutually perpendicular directions.		1) No electrical discontinuity of more than 10 μs. 2) No damage, cracks or looseness of parts.	X	—												
ENVIRONMENTAL CHARACTERISTICS																	
Rapid Change of Temperature	Temperature: -40 → R/T ⁽¹⁾ → +125 → R/T °C Time: 30 → 2 to 3 → 30 → 2 to 3 min for 5 cycles.		1) Insulation resistance: 5000 MΩ MIN. 2) No damage, cracks or looseness of parts.	X	—												
Damp Heat, Steady State	Subjected to a temperature of +40°C, at a humidity of 90 to 95% for 96 hours.		1) Insulation resistance: 50 MΩ MIN. (At high humidity) 2) Insulation resistance: 500 MΩ MIN. (When dry) 3) No damage, cracks or looseness of parts.	X	—												
Corrosion Salt Mist ⁽⁴⁾	Subjected to 5% salt spray for 48 hours.		No heavy corrosion which impairs functionality.	X	—												
Sealing ⁽⁴⁾	Subjected to a depth of 2 m for 14 days. (IPX8 Waterproof)(JIS C 0920:2003)		No water penetration into the connector.	X	—												
Air Tightness ⁽⁴⁾	17.6 kPa of air pressure applied to the inside of the mated connector for 30 seconds.		No air bubbles emitted from the inside of the connector.	X	—												
<table border="1"> <thead> <tr> <th></th> <th>COUNT</th> <th>DESCRIPTION OF REVISIONS</th> <th>DESIGNED</th> <th>CHECKED</th> <th>DATE</th> </tr> </thead> <tbody> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </tbody> </table>							COUNT	DESCRIPTION OF REVISIONS	DESIGNED	CHECKED	DATE						
	COUNT	DESCRIPTION OF REVISIONS	DESIGNED	CHECKED	DATE												
REMARK Notes (1) R/T : Room Temperature (2) Measured contact resistance at the points shown in Fig.1 on the next page. (3) Delaying curve shown in Fig.2 on the next page. (4) Corrosion salt mist, Sealing and Air tightness shall be tested under mated condition with an applicable connector. (5) Operating temperturte range includes the temperature rise by current carrying.			APPROVED	TP. KOMATSU	20220808												
			CHECKED	HY. KOBAYASHI	20220808												
			DESIGNED	TY. SUZUKI	20220805												
			DRAWN	TY. SUZUKI	20220805												
Unless otherwise specified, refer to IEC 60512 (JIS C 5402).																	
Note QT:Qualification Test AT:Assurance Test X:Applicable Test			DRAWING NO.		ELC-118302-04-00												
	SPECIFICATION SHEET		PART NO.	EM30MSD (04)													
	HIROSE ELECTRIC CO., LTD.		CODE NO.	CL0138-0200-0-04	1/2												

FIG.1 Measuring point of contact resistance



[Reference]



Measurement method:

Mated plug and receptacle with 100 mm² cable.

Note:

Derating curve could vary depending on cable type and each measurement even under the same conditions.

Therefore, above data are guidelines and not connector specifications.

Note QT:Qualification Test AT:Assurance Test X:Applicable Test

DRAWING NO.

ELC-118302-04-00

HRS

SPECIFICATION SHEET

PART NO.

EM30MSD (04)

HIROSE ELECTRIC CO., LTD.

CODE NO.

CL0138-0200-0-04



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